

Project Instructions:

Course: Network Security

As part of the academic requirements of this course, you are required to complete a project. You must carefully read and follow all instructions outlined below.

1. Team Formation

You must form a project team consisting of four (4) to five (5) students.

Individual projects are not to be considered.

You are responsible for:

- Forming your team by 14/2/26 ensuring equitable contribution from all team members
- Coordinating tasks such as research, implementation, testing, documentation, and a final presentation (A date for the presentation will be announced later but it will be sometime during the last/second last week of the month of April.)
- Once teams are formed, no changes to team composition will be considered unless absolutely necessary and only with a formal intimation to me.

2. Project Objective and Scope

You must select a project topic related to Network Security that involves both conceptual design and practical implementation. Your project should demonstrate how security mechanisms are applied to protect network communication in realistic scenarios.

Your project must address the fundamental principles of network security, including, but not limited to:

- Confidentiality: Protecting data from unauthorized disclosure using appropriate techniques
- Integrity: Ensuring that data transmitted over the network is not altered or tampered with
- Authentication: Verifying the identities of communicating entities
- Non-Repudiation: Ensuring that parties involved in communication cannot deny their actions,
- Availability: Ensuring reliable access to network services and resistance to denial-of-service attacks
- Access Control and Authorization: Regulating access to network resources based on defined security policies wherever applicable.

You are expected to clearly justify why these security aspects are relevant to your chosen project and how they are implemented.

3. Project Proposal

You must submit a project proposal the same time, you submit your team details. The proposal must include:

- Title of the project
- Problem statement and motivation
- Core objectives of the project
- Overview of the proposed system or solution
- Network architecture and communication model
- Security mechanisms to be implemented
- Tools, technologies, and protocols to be used
- Expected outcomes and evaluation criteria for your implemented functionalities

The project cannot proceed until the proposal has been submitted and approved.

4. Implementation and Network Analysis

You are required to implement a working prototype or simulation that demonstrates your proposed security solution.

A key requirement of this project is the analysis of network communication. You must perform thorough traffic analysis using tools such as Wireshark, along with any other relevant network monitoring or security tools.

Your analysis must include:

- Packet capture and inspection of network traffic
- Identification of normal and abnormal traffic patterns
- Evaluation of how confidentiality, integrity, and authentication are enforced
- Evidence showing how attacks or vulnerabilities are detected or mitigated if relevant
- Discussion of the robustness and limitations of your implementation - very important.

- Screenshots, logs, and observations from Wireshark or similar tools must be included in your report as supporting evidence.

5. Deliverables

You are required to submit the following deliverables:

a) Project Proposal Report

A formal written document stating the proposed project, as mentioned above is to be submitted by the specified deadline.

b) Final Project Report

A comprehensive report documenting:

- Introduction and background
- System design and architecture
- Implementation details
- Network security mechanisms used
- Network traffic analysis and results along with screenshots and outputs.
- Discussion, limitations, and conclusions

The report must be well-structured, technically accurate, and written in formal academic language. It is to be submitted, at least 2 days before the presentation date.

c) Project Presentation

You must deliver a formal presentation summarizing your project work. Include,

- The problem addressed
- The implemented security solution
- Key findings from network analysis
- Challenges faced and lessons learned

All team members are expected to participate in the presentation.

6. Evaluation Criteria

Your project will be evaluated based on:

- Understanding and application of network security principles
- Depth and quality of network communication analysis
- Correct and effective use of tools such as Wireshark
- Technical rigor and originality of the solution
- Quality of documentation and presentation
- Team coordination and contribution

7. Academic Integrity

All work submitted must be original. Any external sources, tools, or references used must be properly cited. Plagiarism will be dealt with according to the institution policies.

8. Innovation and Publication Potential

Projects that demonstrate strong innovative ideas, originality, or significant technical depth may be considered for publication in appropriate academic venues such as conferences, journals, subject to further refinement.

Such consideration will depend on the novelty of the work, quality of implementation, rigor of analysis, and clarity of documentation.